

ANPR SOFTWARE PROJECT

Phase 1 – Requirements & Implementation Plan

This document converts the discussed requirement into a practical Phase 1 implementation plan. The proposed solution will use normal IP surveillance cameras connected to a local edge system. The edge system will handle the AI processing, number-plate recognition, event processing and communication with the web application.

1. Phase 1 Objective

Build an on-premises vehicle and number-plate monitoring system using **normal IP surveillance cameras** instead of dedicated ANPR cameras. The cameras will provide RTSP video streams and the AI/computer-vision processing will be handled by the edge application.

CCTV / RTSP Camera → Edge System → Vehicle Detection → Number Plate Detection → OCR → Registered Vehicle Matching → Rules/Alerts → Database → Web UI → Excel Report

2. Edge System

The edge system will be the local PC/server connected to the CCTV network. It will receive the camera streams, run the required AI models, process the number plates and generate events locally.

Component	Purpose
RTSP camera input	Receive live video from normal surveillance cameras.
AI model 1 – Vehicle detection	Detect vehicles such as cars, bikes and other required vehicle classes.
AI model 2 – Number plate detection	Locate the number plate inside the vehicle/frame.
OCR	Read the characters from the detected number plate.
Vehicle matching	Compare the detected plate with the client's registered vehicle list.
Rules engine	Classify resident, outsider, unknown, whitelist/blacklist and alerts.
Event storage	Save plate, time, camera, image, confidence and classification.

3. AI / Computer Vision Requirement

Because dedicated ANPR cameras are not being used, the AI work becomes an important part of the system. The software will need to identify the vehicle, find the number plate and read the characters from it.

This means additional AI/computer-vision work is required for vehicle detection and number-plate detection, followed by OCR to read the detected plate.

- **Vehicle Detection Model:** Detect the vehicle and provide a bounding box/confidence.
- **Number Plate Detection Model:** Detect the number plate region from the vehicle/frame.
- **OCR:** Convert the plate image into readable text, for example UP32AB1234.
- **Quality checks:** Filter very poor detections before creating a final event.

The exact model, OCR engine and tuning will depend on the camera footage and sample images supplied by the client. Camera placement, lighting, distance, motion and plate visibility will affect recognition performance.

4. Registered Vehicle / Resident Matching

The client will provide the list of registered vehicles. This list will be imported into the vehicle database and used to classify detected vehicles.

Detected event	System result
Plate found in registered list	Resident / Registered Vehicle
Plate not found	Outsider / Unknown Vehicle
Plate in blacklist	Blacklist Alert
Plate in whitelist	Whitelist / Authorized Vehicle

For a registered vehicle, the event can show vehicle number, owner/resident name, vehicle type and registration status, depending on the information supplied by the client.

5. Web UI – What We Will Build

Screen	Main information / actions
Login	User login and Admin / Operator access.
Dashboard	Today's vehicle count, entries/exits, recent events, alerts and camera status.
Live Events	Live/latest detections with plate number, vehicle image, camera, time, direction and status.
Vehicle Database	Registered vehicles, owner/resident details, vehicle type, status and validity.
Vehicle History	Complete movement/detection history for a selected vehicle/plate.
Alerts	Unknown, blacklist and other configured alerts with acknowledgement.
Cameras	Camera name, IP/RTSP details, location/gate, direction and status.
Reports	Date-wise, vehicle-wise, camera-wise and IN/OUT reports with Excel export.
Users	Admin and Operator management.
Settings	Basic site, storage, retention and system settings.

6. Excel Report

The Excel report should contain enough information to understand what happened for each detected vehicle. Suggested columns are:

Field	Example / purpose
Date & Time	When the vehicle was detected.
Plate Number	OCR result, e.g. UP32AB1234.
Vehicle Type	Car / bike / other detected or registered type.
Camera	Camera name/location.
Direction	IN / OUT, where configured.
Vehicle Status	Registered / Outsider / Unknown / Whitelist / Blacklist.
Resident / Owner	Name from registered vehicle list, when matched.
Confidence	Detection/OCR confidence where available.
Image Reference	Saved image/event reference.

7. Alerts & SMTP

The system can generate alerts for events such as blacklisted vehicles, unknown/outsider vehicles or other configured conditions.

SMTP email alerts: If email notification is required, SMTP will be used to send alerts to configured recipients. The client will need to provide the SMTP server, port, security method and sender account details.

8. Event Processing Flow

1. CCTV sends RTSP video → 2. Edge reads frames → 3. Vehicle model detects vehicle → 4. Plate model detects plate → 5. OCR reads plate → 6. Plate is matched with registered vehicle list → 7. Rules classify event → 8. Event/image is stored → 9. UI updates → 10. Alert/Excel data becomes available.

9. Data / Information Required From Client

- **CCTV camera make/model and RTSP access details** at the time of testing and deployment.
- Camera locations and IN/OUT configuration, if applicable.
- Sample live/recorded footage or camera snapshots for AI testing.
- Registered vehicle list with plate number and resident/owner information.
- Required vehicle categories to recognize.
- Whitelist/blacklist rules and alert conditions.
- SMTP details and recipients, if email alerts are required.
- Required Excel report fields/format.
- **Approximate number of cameras expected to be connected to one edge system.**
- Server/edge PC details and available GPU have already been received.

10. Phase 1 Scope

Phase 1 will focus on the core edge-based ANPR workflow: CCTV/RTSP connectivity, AI vehicle detection, number-plate detection, OCR, registered vehicle matching, rules and alerts, database/event storage, web UI and Excel reporting.

Gate/barrier integration is not included in Phase 1 and can be discussed separately if required later.

11. Phase 1 Deliverable

At the end of Phase 1, the target is a working on-premises edge-based ANPR application connected to the provided surveillance camera(s), with AI vehicle detection, number-plate detection, OCR, registered vehicle matching, event/alert handling, web UI and Excel reporting.

Important: Before finalizing the implementation details, the camera count per edge, camera models/RTSP availability, sample footage and registered vehicle data should be confirmed with the client.